

Product Requirements Document: Chemo Companion

Metadata	Details
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Product Owner	Team Cura
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1. Executive Summary

Core Function: An AI-driven oncology assistant and ecosystem designed for 24/7 symptom monitoring, caregiver support, and community integration.

Technical Foundation:

- *Multi-Agent Orchestration:* A specialized workflow involving Planner, Lookup, Relevance, Browser, Writer, Hallucination-Check, Reviewer, and Citation agents to ensure high-accuracy responses.
- *Hybrid GraphRAG:* Grounding AI responses in a combination of curated scientific documents (extracted via LangExtract) and real-time browser searches from verified domains (e.g., cancer.org).
- *Local LLM Integration:* Utilization of SEA-LION and MERaLiON for culturally nuanced communication and multimodal sentiment analysis.
- *Safety Framework:* Implementation of Guardrails AI and hard-coded "Red Flag" detection for clinical escalation.

Strategic Goals:

- *Continuous Monitoring:* Managing physical symptoms, medication adherence, and appointment scheduling.
- *Clinical Reporting:* Generating data-driven history reports for doctors based on patient logs and wearable device metrics.
- *Community Enablement:* Using Knowledge Graphs to match patients with peer groups and relevant activities at Active Ageing Centres (AACs).

2. Problem Statement

The current oncology experience in Singapore suffers from three primary gaps:

- *The Information Gap:* Patients and caregivers often lack 24/7 access to reliable, personalized medical information between clinic visits. Generic internet searches often lead to misinformation or unnecessary anxiety.
- *The Monitoring Gap:* Side effects and vital signs are often not recorded systematically between appointments. This results in "recall bias" during doctor consultations, where patients forget to report significant but transient symptoms.
- *The Support Gap:* Caregivers face high mental strain and fatigue without a dedicated support system. Simultaneously, patients face social isolation despite the presence of community resources like AACs, because there is no mechanism to match their current medical state with safe, appropriate activities.

3. User Personas

Persona	Primary Needs
The Patient	Needs immediate answers to symptom concerns, medication reminders, and a way to track their condition for their doctor.
The Caregiver/Family Member	Needs help managing the patient's schedule and emotional support/intervention when fatigue is high.
The Oncologist	Needs a summarized, accurate report of the patient's health metrics and side effects from the past week to make informed clinical decisions.

4. User Journeys

This section describes how a user interacts with the system and how the background agents process that interaction.

4.1. Monitoring

- Journey 1: Disease-Specific Doubt Resolution (Use Case 1a)
 - *User Action:* A patient experiences a sudden side effect (e.g., "My fingernails are turning dark") and asks the assistant for an explanation.
 - *The Multi-Agent Workflow:* The Planner identifies a medical query. The Lookup Agent searches the internal GraphRAG (local documents). If the answer is not specific enough, the Relevance Agent triggers the Browser Agent to search verified domains (e.g., cancer.org).
 - *The Result:* The patient receives a grounded, cited explanation: "This is a common side effect of your medication (Docetaxel) called hyperpigmentation. It is usually temporary. I have logged this for your next doctor's visit."
- Journey 2: Proactive Health Suggestions (Use Case 1b)
 - *User Action:* The system detects a significant drop in the patient's average sleep and an increase in resting heart rate via wearable data.
 - *The Multi-Agent Workflow:* The Planner recognizes a health metric anomaly. It directs the Lookup Agent to find recovery advice for the patient's current chemo phase.
 - *The Result:* The system sends a proactive notification: "Your sleep quality has been low for three nights. We suggest avoiding caffeine after 2 PM and trying the 5-minute breathing exercise in the 'Wellness' tab."
- Journey 3: Daily Check-in & Management (Use Case 1c)
 - *User Action:* The patient opens the app for their scheduled morning check-in.
 - *The Multi-Agent Workflow:* The system asks structured questions about mental state, physical symptoms, and medication. It checks the digital calendar for upcoming appointments.
 - *The Result:* The system logs the responses and provides a summary: "Meds logged. Reminder: You have a blood test at 2 PM today. I have sent a notification to your transport contact."

4.2. Reporting

- Journey 4: Weekly Patient History Sharing (Use Case 2a)
 - *User Action:* The patient prepares for a weekly consultation with their oncologist.

- *The Multi-Agent Workflow:* The Writer Agent aggregates all data from the past 7 days, including side-effect logs, medication adherence, and wearable metrics (steps/heart rate).
- *The Result:* A structured report is generated and shared with the doctor, providing an objective view of the "between-visit" period to inform better clinical decisions.
- *Note:* There are templates for this.

4.3. Enabling Community

- Journey 5: Peer & AAC Matching (Use Case 3a)
 - *User Action:* The patient expresses feeling lonely or "stuck at home."
 - *The Multi-Agent Workflow:* The Lookup Agent scans the Knowledge Graph to find patients with similar treatment profiles and checks the local AAC (Active Ageing Centre) schedule for safe, relevant activities.
 - *The Result:* The system suggests: "There is a 'Low-Impact Taichi' session at the Bedok AAC tomorrow. Three other oncology patients from our community are attending. Would you like to join them?"
- Journey 6: Patient Progress Visualization (Use Case 3b)
 - *User Action:* The patient views their "My Journey" tab to see their treatment status.
 - *The Multi-Agent Workflow:* The system pulls historical data from the Knowledge Graph, comparing the current cycle to previous ones.
 - *The Result:* The patient sees a visual dashboard: "You have successfully completed 50% of your treatment plan. Your energy levels this cycle are 15% higher than the last." This empowers the patient with a sense of forward momentum.

5. Functional Requirements

5.1. Umbrella 1: Monitoring (Proactive & Reactive)

- *FR 1: Multimodal Daily Check-ins*
 - The system shall initiate a daily check-in via text or voice (using MERaLiON) to capture mental state, medication adherence, and physical symptoms.
 - Agent Logic: The Planner Agent shall trigger specific sub-flows (e.g., if a user mentions pain, it initiates a "Symptom Grading" flow).
- *FR 2: Grounded Q&A Assistant*
 - The system shall provide answers to oncology-related queries grounded in a GraphRAG knowledge base.
 - Agent Logic: The Lookup Agent shall query local docs (via LangExtract); if the relevance score is low, the Browser Agent shall fetch data from cancer.org.
- *FR 3: Appointment & Med Management*
 - The system shall allow users to add appointments and medication schedules via natural language.
 - The system shall provide proactive reminders 24 hours and 2 hours before events.

5.2 Umbrella 2: Reporting (Clinical Visibility)

- *FR 4: Wearable Data Integration*

- The system shall ingest step counts and heart rate data from wearable devices (simulated or real) to identify trends.
 - Agent Logic: The Relevance Agent shall flag anomalies (e.g., significant drop in activity) to the patient.
- *FR 5: Automated Physician Summary*
 - The system shall generate a structured PDF report summarizing the last 7 days of patient data (symptoms, vitals, adherence).
 - Agent Logic: The Writer Agent shall synthesize unstructured chat logs and structured metrics into a professional clinical tone.

5.3 Umbrella 3: Enabling Community (Social Integration)

- *FR 6: Intelligent AAC & Peer Matching*
 - The system shall match patients with similar conditions and interests using the Knowledge Graph.
 - The system shall recommend specific Active Ageing Centre (AAC) activities based on the patient's current energy levels and clinical state.
- *FR 7: Progress Visualization*
 - The system shall display a "Recovery Roadmap" that tracks treatment cycles completed and mobility trends.
 - Agent Logic: The Writer Agent shall generate a personalized "weekly win" message to boost morale.

5.4 Safety & Guardrails

- *FR 8: "Red Flag" Word Detection*
 - The system shall use a hard-coded interceptor and Guardrails AI to detect life-threatening keywords (e.g., "shortness of breath", "fever > 38.5").
 - Action: If detected, the system shall immediately stop the agentic workflow and provide a one-click button to call the hospital or 995.
- *FR 9: Citation Enforcement*
 - Every medical answer must include a citation from the Citation Agent linking back to a specific source or URL.

6. Non-Functional Requirements (NFRs)

- *Trust & Accuracy:* AI responses must be grounded in the provided Knowledge Graph or verified domains. The "Hallucination Check" agent must have a threshold of 100% for medical citations.
- *Latency:* For the "Assistant" feel, the Planner Agent should provide an initial "Thinking..." response within 2 seconds to manage user expectations during complex multi-agent calls.
- *Cultural Resonance:* The UI and tone (via SEA-LION) must support Singaporean linguistic nuances.

7. Proposed System Architecture

This architecture utilizes a Hybrid GraphRAG approach with a Manager-Worker agentic pattern.

7.1. Data Layer

- *Knowledge Base*: Structured data extracted via LangExtract from clinical PDFs stored in a Vector DB (e.g., Chroma or Pinecone) with Metadata mapping to represent the "Graph."
- *External Sources*: Real-time search tool restricted to site:cancer.org, site:nuhs.edu.sg, etc.

7.2. Multi Agent Orchestration for chatbot (can add another agent for scheduling)

We can use a Sequential & Branching Workflow:

- *Input Handler*: Captures Text/Voice (MERaLiON).
- *The Planner (Orchestrator)*: Analyzes Persona + Intent. Decides the branch (Monitoring vs. Community vs. Admin).
- *The Lookup Agent*: Queries the internal Knowledge Graph.
- *The Relevance Agent*: A "judge" that decides if internal data is sufficient. If NO then calls Browser Agent (which looks up only in the external sources list we specify).
- *The Writer Agent*: Synthesizes data using SEA-LION.
- *The Safety Layer (Guardrails AI)*: Runs parallel to all agents to detect "Red Flags" and triggers the hard-coded "Call 995" interrupt.
- *The Reviewer & Citation Agents*: Final check for hallucinations and appending of source links.

7.3. Front-End, Check-ins, Appointment Scheduling, AACs activities and Reporting (according to framework)

- Reporting patient history in specific formats
- Smooth seamless and easy to 'emotionally connect' to UI-UX.

8. Technical Constraints

- *Model Availability*: Dependency on the SEA-LION and MERaLiON APIs/local instances being performant for real-time chat.
- *Context Window*: Multi-agent loops can be token-heavy. The Planner Agent must prune the history to keep the most relevant medical context or cache FAQs (can be an add on much much later)
- *Connectivity*: The Browser Agent requires active internet access; if offline, the system must gracefully degrade to "Internal Knowledge Only" mode.
- *Interoperability*: For the innovation challenge, we should assume mock APIs for HealthHub and AAC schedules.

9. Success Metrics

- Product Metrics:
 - *Information Reliability*: % of queries successfully cited vs. % of queries escalated to 'Consult a Doctor'.
 - *Task Success Rate*: % of appointments/med reminders successfully scheduled via natural language.
 - *Caregiver Offload*: Reduction in "manual" clicks for the caregiver (calculating the number of tasks handled by the agent).
- Technical Metrics:

- *Agent Efficiency*: Average time taken from "User Query" to "Final Response" across the 8-agent loop.
- *RAG Fidelity*: Accuracy of LangExtract in mapping PDF data to the correct Graph nodes